Supriya Pingle (a1944342)
BUSANA 7001 Visual and Predictive Analytics Assignment
Task 1 – Visual Slides

Formatted Table: Key Financial Metrics by State (Top 15 States by ROA)

The table summarizes the mean financial characteristics for firms in the 15 states with the highest Return on Assets (ROA). Large total asset (AT) bases for states like Texas (TX), California (CA), and New York (NY), where the biggest portions of highly capitalized firms are headquartered obviously support this assertion. Smaller states (ex: NE, SC) indicate firms with smaller scale of operations. Variations in profitability (ROA and profit margin) space varies widely. Some states have higher averages, and others are very close to breakeven, which characterizes competitive pressures of profitability.

STATE	ROA	AT	SALE	profit_margin	MB	ret	Frequency
QC	0.33	3.911	7.7	0.25	2.2	0.8	1
NE	0.33	7.676	18	0.12	1.2	0.12	1
SC	0.32	23.399	57	0.12	1.6	1.4	1
NC	0.32	35.836	93	0.21	3.4	-0.29	2
NJ	0.32	56.677	111	0.13	1.4	0.77	2
CT	0.33	86.155	80	0.01	1.2	0.17	2
TX	0.33	119.626	218	0.14	1.3	0.25	8
WA	0.33	143.567	219	0.09	2.2	1.4	1
IL	0.34	201.387	190	0.1	1.4	1.5	1
MD	0.34	322.818	255	0.15	2	0.94	3
PA	0.33	330.772	509	0.09	2.1	1	1
IN	0.32	629.051	940	0.23	2.4	0.33	1
NY	0.33	817.321	353	0.21	1.2	-0.17	3
CO	0.33	1191.753	871	0.13	1.3	0.31	4
CA	0.33	1391.048	834	0.14	1.7	0.04	6

Figure 1

Histogram of Profit Margin

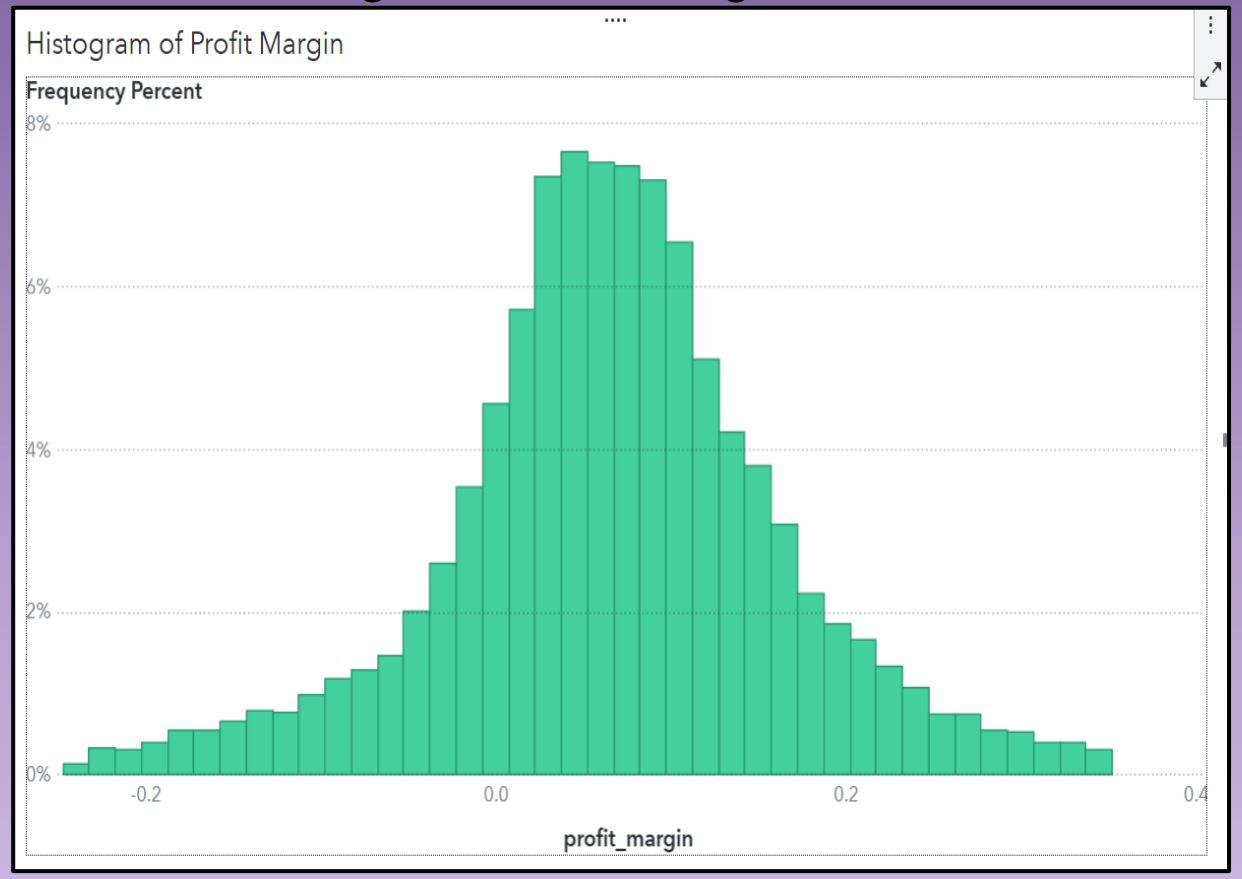

Figure 2

The histogram indicates that the profit margins of most firm's cluster around zero. Thus, many firm operate close to breakeven. The shape of the distribution is approximately bell shaped but skewed with respect to profit margin, clearly evident with several firms reporting negative margins indicating there are firms also experiencing losses.

Histogram of Total Assets (AT)

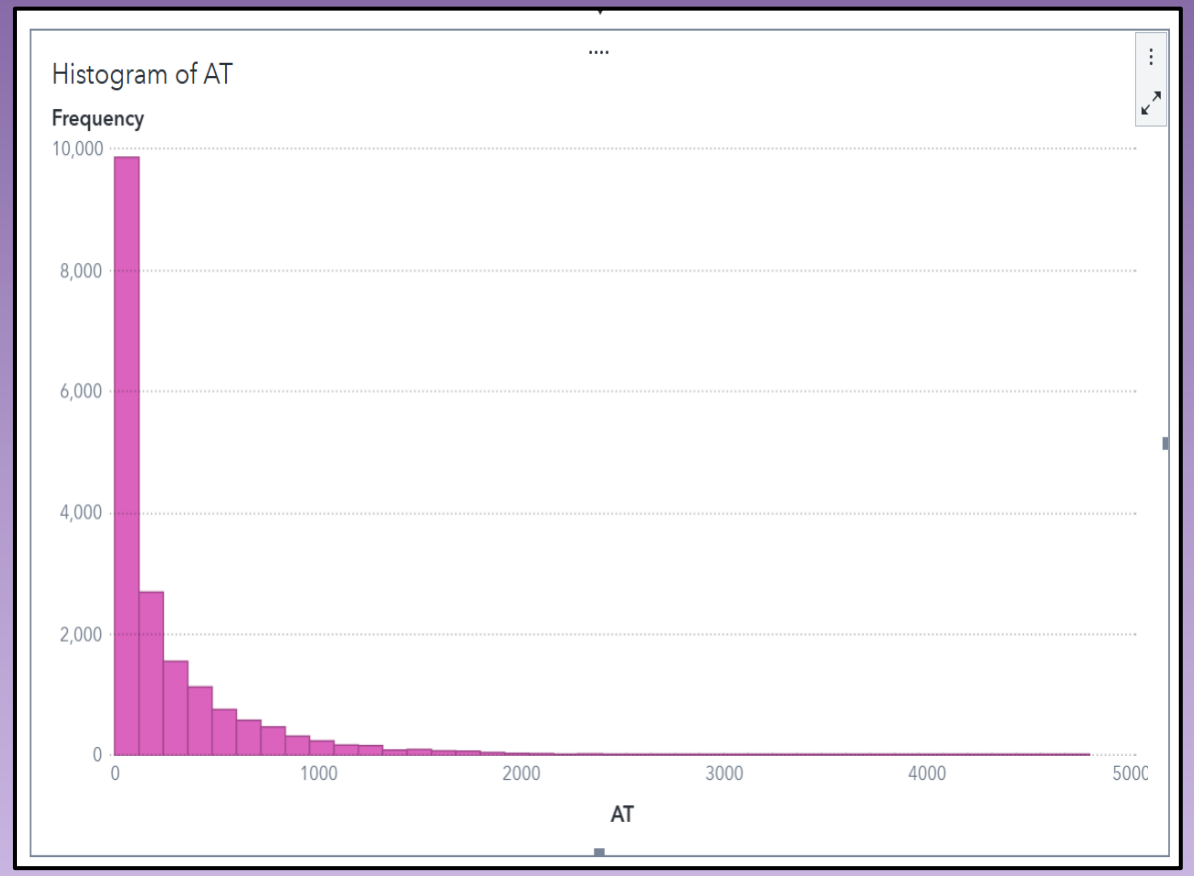

Figure 3

The histogram of total assets indicates an extremely right-skewed distribution. Most firms in the sample are quite small in nature while a handful of very large firms represent the overwhelming portion of the observations. The skewness of the distribution confirms the use of log transformations later in the regression analysis.

Pie Chart of Total Assets (AT) by Company

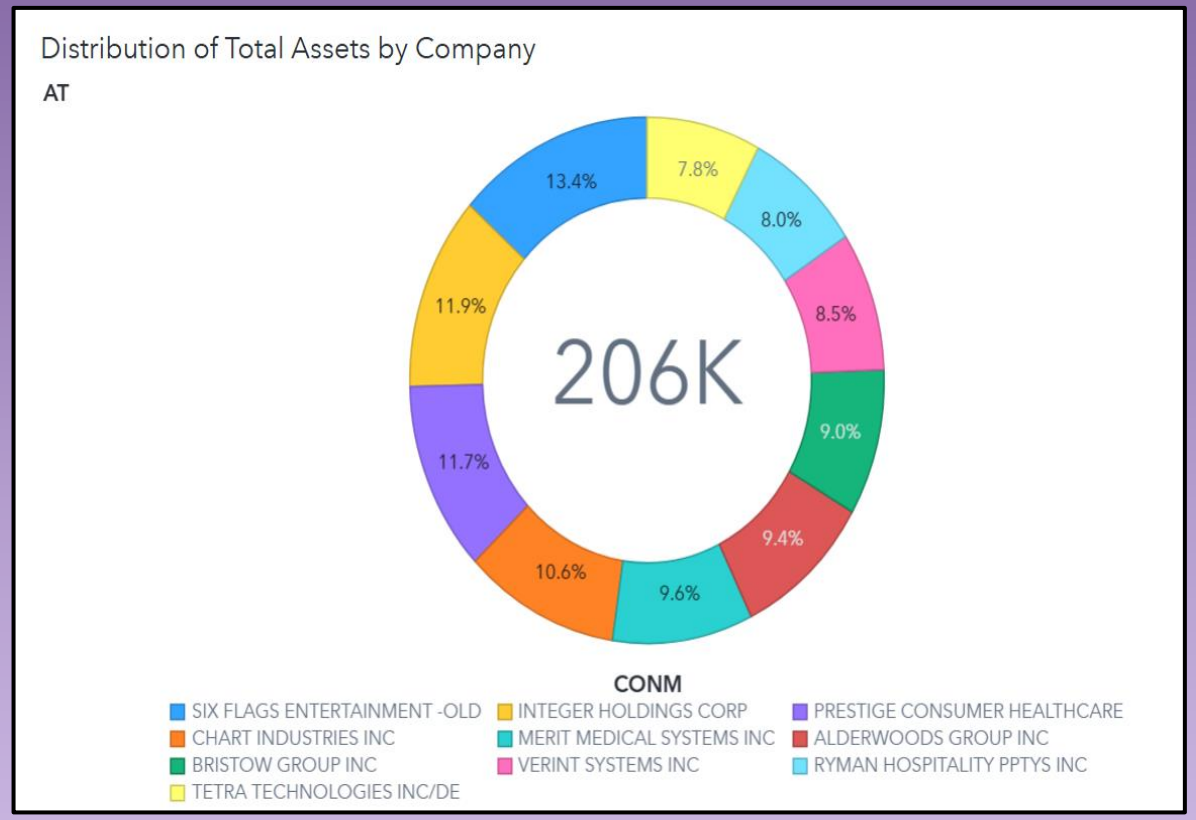

Figure 4

The pie chart shows how firms' total assets are allocated. There are a few firms that dominate assets with a fairly significant number of firms holding very little in assets; this is helpful just in affirming the degree of concentration and reflects the distribution of firm sizes in the dataset.

Bar Chart of Asset Growth by Company

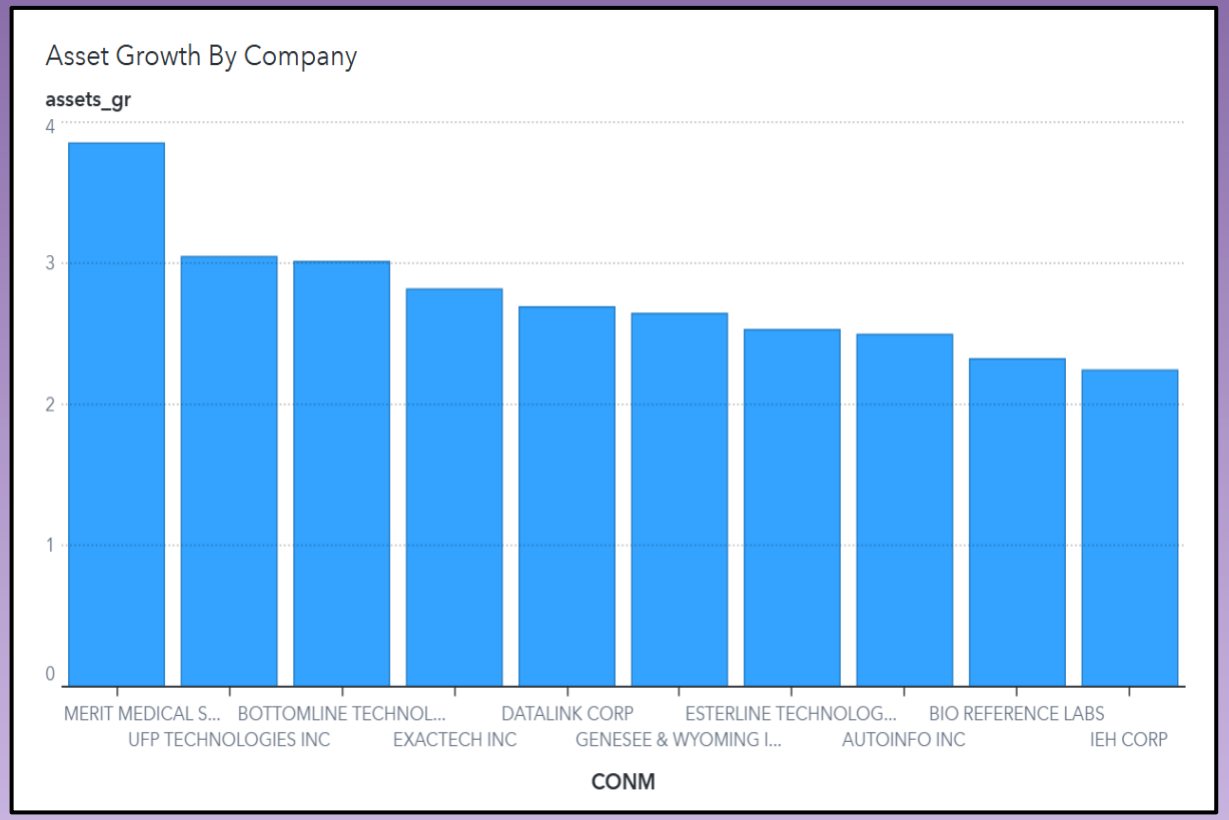

Figure 5

The bar chart highlights asset growth in a firm. A couple of firms stand out, such as Merit Medical and UFP Technologies, with far more growth than the rest of the group. The other firms had very little or negative growth. Firms with high growth rates may attract acquisition interest, as there may be downside risk to further negative or growth, and the weaker companies may even become targets for acquisition.

Scatter Plot of Total Assets vs Sales

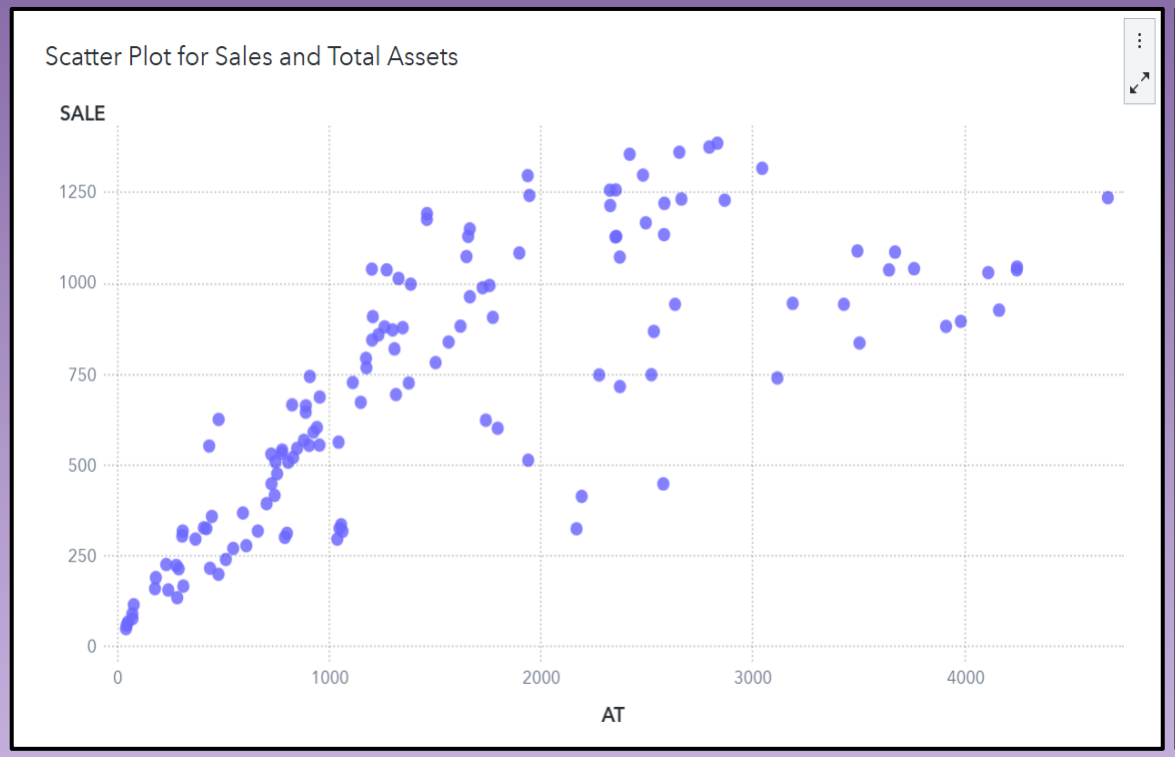

Figure 6

The scatter plot demonstrates an unambiguous positive correlation between total assets and sales. Larger firms have more total assets meaning greater revenue and confirms the link between size or scale and performance. There are a few firms with assets, and sales relatively lower than others, which could indicate inefficiencies, and possible candidates for M&A activity.

Heat Map – Total Assets by State

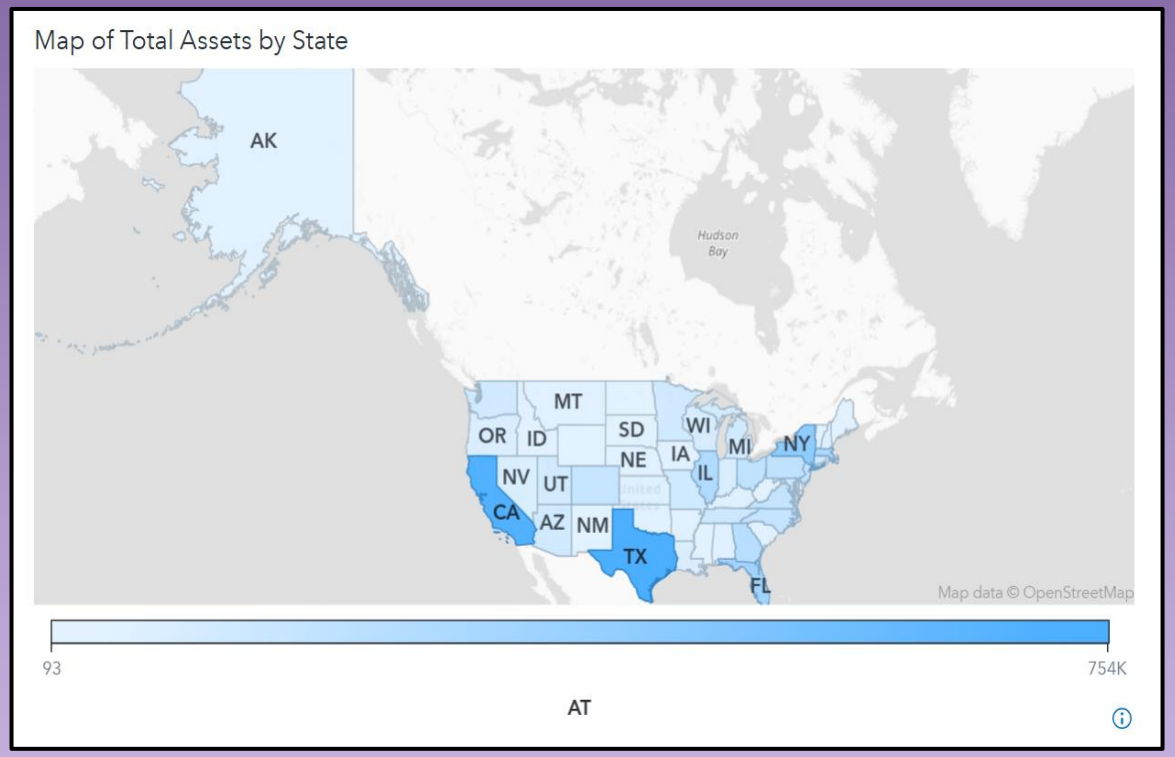

Figure 7

The heat map emphasizes geographic distribution of total assets in each of the states showing that Texas and California have a disproportionate amount of total assets shown in size and followers New York and Florida. The cluster of states suggests certain states are more concentrated corporate hubs, which may create higher total asset pathways for more M&A activity.

Conclusion – Descriptive Insights

- The distribution of size in firms is very skewed; while most firms are relatively small, a few firms tend to be very large.
- Firm profitability is inconsistent; many firms are unprofitable, and many firms are close to breakeven.
- Asset growth is variable; while some firms are experiencing strong growth, others are declining.
- Firms are valued at their book value, which is consistent for valuation multiples (MB) clustering to 1-2.
- There was some variation at the state level, with Texas, California, and New York commanding the largest amount of total assets (and therefore could potentially be M&A hotspots).
- To some extent, differences in industry composition across the states help explain differences in size and profitability.
- According to geographic clustering, there are some areas that will have more M&A activity and attract a higher concentration of businesses.
- The theory shows descriptive trends, that the M&A target is typically a smaller, less profitable firm, or growth firm.